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Economic Evaluation

Consolidated Health Economic Evaluation Reporting Standards - Value of Information (CHEERS-VOI): Explanation and Elaboration



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ABSTRACT

Objectives: Although the ISPOR Value of Information (VOI) Task Force's reports outline VOI concepts and provide good-practice recommendations, there is no guidance for reporting VOI analyses. VOI analyses are usually performed alongside economic evaluations for which the Consolidated Health Economic Evaluation Reporting Standards (CHEERS) 2022 Statement provides reporting guidelines. Thus, we developed the CHEERS-VOI checklist to provide reporting guidance and checklist to support the transparent, reproducible, and high-quality reporting of VOI analyses.

Methods: A comprehensive literature review generated a list of 26 candidate reporting items. These candidate items underwent a Delphi procedure with Delphi participants through 3 survey rounds. Participants rated each item on a 9-point Likert scale to indicate its relevance when reporting the minimal, essential information about VOI methods and provided comments. The Delphi results were reviewed at 2-day consensus meetings and the checklist was finalized using anonymous voting.

Results: We had 30, 25, and 24 Delphi respondents in rounds 1, 2, and 3, respectively. After incorporating revisions recommended by the Delphi participants, all 26 candidate items proceeded to the 2-day consensus meetings. The final CHEERS-VOI checklist includes all CHEERS items, but 7 items require elaboration when reporting VOI. Further, 6 new items were added to report information relevant only to VOI (eg, VOI methods applied).

Conclusions: The CHEERS-VOI checklist should be used when a VOI analysis is performed alongside economic evaluations. The CHEERS-VOI checklist will help decision makers, analysts and peer reviewers in the assessment and interpretation of VOI analyses and thereby increase transparency and rigor in decision making.

Keywords: CHEERS checklist, health economic evaluation, reporting guideline, value of information.

VALUE HEALTH. 2023; 26(10):1461–1473

Introduction

In healthcare, decisions made by policymakers and payers are often supported by economic evaluations (EEs), which compare the health outcomes and costs associated with alternative decision options.^{1,2} In common with all decision processes, EEs use imperfect information, meaning that the impact of the resulting uncertainty on the decision and the potential consequences of making suboptimal decisions should be evaluated.^{3–5} This can be achieved using a value of information (VOI) analysis to quantify the expected value of collecting additional information before making the decision.^{6–8} VOI can be used to perform sensitivity analyses,⁹ prioritize future research,¹⁰ and determine optimal future study designs.^{11,12}

The Consolidated Health Economic Evaluation Reporting Standards (CHEERS) 2022 statement provides a minimum set of items required for reporting EE¹³ and has had substantial impact

on EE reporting.¹⁴ As EEs have become more widely used, so have VOI analyses.^{15,16} Although previous work has discussed elements that affect VOI estimates and provided summaries of the methods and concepts,^{17,18} no checklist on the minimum, essential information to be reported from a VOI analysis exists.¹⁹ Thus, it can be challenging to assess, interpret, and compare VOI analyses because key information may be missing from published studies.

The present work develops the CHEERS-VOI checklist, which adapts the CHEERS statement to specify the minimum, essential information for reporting an EE that includes a VOI analysis. When a VOI analysis is performed alongside EE, the CHEERS-VOI checklist should be used instead of the checklist provided by the CHEERS statement. The CHEERS-VOI checklist has the potential to improve the transparency, reproducibility, and rigor of VOI reporting. We describe the development of the CHEERS-VOI checklist, including an elaboration on new items and those that overlapped with the CHEERS statement but needed

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a modification, and clarify when the CHEERS-VOI should be used. The primary audiences for the CHEERS-VOI checklist are decision analysts, decision makers, and other stakeholders who perform or use the results of VOI analysis.

Approach

The development of the CHEERS-VOI checklist followed recommendations from the EQUATOR (Enhancing the Quality and Transparency of Health Research) Network and the methodology used to develop similar guidelines.^{13,20} It was led by a steering committee, which consisted of 10 members (NK, AS, MD, SG, JG, DH, HK, CR, ECFW, and AH), including some members of the Collaborative Network on Value of Information (ConVOI; <https://www.convoi-group.org/>), other experts in VOI, EE, and guidelines development, and the researchers who led the development of the CHEERS statement. We started the checklist development with updating a previous literature review¹⁶ to extract a list of 26 candidate reporting items, which were reviewed, edited, and confirmed through discussion within ConVOI. We identified items that are relevant to VOI analysis in general, as well as items that are only relevant to specific VOI measures. The definitions of the 4 key VOI measures, including expected value of perfect information (EVPI), expected value of partial perfect information (EVPPI) and expected value of sample information (EVS), and expected net benefit of sampling (ENBS) are provided in Table 1.^{3,9,11,12}

Delphi Study

The candidate reporting items (N = 26) were reviewed through an international Delphi study.²¹ The CHEERS-VOI steering committee nominated individuals who (1) had knowledge of or experience in VOI analysis, or a combination of both, and (2) represented a range of fields of EE and geographic locations. Email invitations were sent to nominated individuals, which they were encouraged to cascade to relevant individuals in their networks. All registrants who self-reported familiarity or strong expertise with VOI were eligible to participate. To maximize the recruitment, we used a snowball sampling method.²² A total of 30 individuals registered in our Delphi study, of whom 74% work in the academic sector, 13% work in the governmental sector, and 13% work in the private sector. All registrants who wished to be acknowledged are listed in the acknowledgments section. All 30 registrants provided their responses in round 1, 25 registrants provided their responses in round 2, and 24 in round 3. In this 3-round Delphi study, each item was scored using a 9-point Likert

scale²³ and participants were invited to provide comments regarding each item's description and indicate if they lacked confidence in their rating. Although the Delphi participants were not asked to provide suggestion for new candidates for reporting items, they had the opportunity to suggest changes in the proposed items (merging, division, or revision) and inclusion of additional information to be reported. After each round, 2 authors (AH and AS) collated comments and initiated the next round with information about item ranks, median scores, interpercentile ranges, and summarized aggregate comments, which were reviewed by the steering committee.

All items with a mean score >7 (or <4) after round 2 were included (excluded) in the consensus meeting, whereas all remaining items proceeded to round 3. New or combined items in round 2 also proceeded to round 3. Participants only rated the remaining items in round 3 and those with a final mean score of 4 or less were excluded.

Consensus Meeting

Members of the steering committee (NK, AH, ECFW, HK, JG, SG, CR) participated in 24-h virtual consensus meetings to determine the final CHEERS-VOI checklist. Before the meeting, a summary of all the items, including their mean scores, comments, and summarized aggregate comments, was provided to attendees. Moderated round table discussions followed a presentation of the Delphi results for each item to finalize the wording and title. After up to 20 minutes, attendees voted anonymously to either include, exclude, or merge each item. Attendees could also be "Unsure." A final decision on each item required at least 70% of attendees to vote "Include" or "Exclude." Any items that did not reach consensus after the first round were discussed and voted on again until consensus was reached.

CHEERS-VOI Reporting Standards

Scope

The CHEERS-VOI checklist is intended to be applied when performing a VOI analysis alongside EE. Although EEs are predominantly performed using decision-analytic models, many EEs are also performed alongside a clinical trial.²⁴ When a VOI analysis is performed alongside a trial-based EE, the authors should use the CHEERS-VOI checklist and exclude the items that are only relevant to model-based EEs and VOI. When structuring the present adaptation of the CHEERS statement, we followed the

Table 1. Definition of key value of information measures.

Measure	Definition	Source
EVPI	The expected value of perfect information calculates the expected value of eliminating uncertainty in all parameters underlying a decision-analytic model.	3
EVPPI	The expected value of partial perfect (or sometimes "perfect parameter") information calculates the expected value of eliminating uncertainty in a subset of the model parameters underlying a decision-analytic model.	9
EVS	The expected value of sample information estimates the expected value of reducing uncertainty in the parameters underlying a decision-analytic model through a proposed study with specific target outcomes and sample size.	11
ENBS	The expected net benefit of sampling (sometimes ENGS; expected net gain of sampling) calculates the net value of undertaking a study with a specific design and is defined as the difference between the EVS and the cost of undertaking the proposed study.	12

ENBS indicates expected net benefit of sampling; EVPI, expected value of perfect information; EVPPI, expected value of partial perfect information; EVS, expected value of sample information.

PRISMA network meta-analysis extension.²⁵ More specifically, the CHEERS-VOI checklist incorporates the items included in the CHEERS statement but with modification to some items for reporting of VOI analyses and addition of several new items. When VOI analysis is performed alongside EE, the CHEERS-VOI checklist should be used for appropriate reporting instead of the checklist provided by the CHEERS statement. Table 2 presents the CHEERS-VOI checklist, including an explanation of each item and a space to highlight where the items are reported. The modifications of existing items included in the checklist provided by the CHEERS statement are indicated with text in italics, whereas the new items were numbered starting with 1 and proceeded with letter “S.” Although each new item was placed in the order that was assessed as the most logical in relation to the core CHEERS statement, the CHEERS-VOI checklist should not be perceived as a prescription of an order in which the items should be addressed. In other words, the authors are free to choose their preferred order of reporting the checklist items but they are recommended to indicate the section heading and paragraph number of the article where each item can be found.¹³

Application of the CHEERS-VOI Checklist

We present the application of the CHEERS-VOI checklist in line with the presentation used in the CHEERS statement.¹³ More specifically, for each item included in the CHEERS-VOI checklist that was either modified or new compared with the items included in the core checklist provided in the CHEERS statement, we provide an explanation, as well as an example in form of a quotation from the previous study that correctly reported the explained item. Of note, any citations and numerations of tables and figures were removed from the quotations and can be found in the articles that the quotations were taken from. The explanation and examples of the remaining, unchanged items are provided in the CHEERS statement report.¹³

Modified Items

The items presented below were included in the core checklist provided in the CHEERS statement but needed to be modified when a VOI analysis is reported alongside EE.

Item 5

Study population: Describe characteristics of the study population (such as age range, demographics, socioeconomic, or clinical characteristics). *If population-level VOI measures are estimated, describe and justify how the population who benefits from the research was defined and how the incidence and/or prevalence were derived.*

Explanation: VOI measures usually estimate the value provided to each person who would be affected by the decision. The population-level VOI estimates the total VOI, summed over all the affected individuals and expected decision time horizon, and it can support research prioritization.¹² The characteristics of this population must be defined alongside the methods used to determine its size. One common method for estimating the population size is to define the yearly number of individuals affected by the decision (yearly incidence) and multiply by the number of years before the decision space changes (decision time horizon), for example, with the development of a new treatment, discounting appropriately.²⁶ It is important not to confuse the time horizon of a decision model with the decision time horizon of a VOI analysis. Depending on the nature of the disease and treatment, the current set of prevalent patients may

also be able to benefit from the research, and these may be added to the incident population in the first year. If this method is used, it is recommended to clarify how the yearly incidence and/or prevalence is estimated and justify the time horizon assumed for the information collected by research to be useful (Box 1²⁷).

BOX 1. Example of item 5: population to benefit from research.²⁷

“We extrapolated EVPI from an individual- to a population-level estimate based on the total number of patients that could benefit from treatment over a 5-year decision lifetime [citation provided]. A 5-year decision time horizon was adopted as this corresponds with when generic versions of febuxostat may become available based on the expiration of key patents [citation provided]. Approximately 30% of current US prevalent gout cases ($n = 8,300,000 \times 0.30$) and 30% of incident gout cases (annual incident cases $n = 199,360 \times 0.30$) were assumed to be candidates for urate-lowering therapy [citation provided]. The benefit of information from additional research was discounted by 3% [citation provided].”

EVPI indicates expected value of perfect information; US, United States.

Item 10

Discount rate: Report the discount rate(s) *used in the economic evaluation and to estimate the population-level VOI analysis and justify the choice.*

Explanation: The value of information generated by research should be discounted over the assumed time horizon to put less weight on the decisions informed in the more distant future and more weight on the decisions informed in the near future¹⁸ (Box 2²⁸).

BOX 2. Example of item 10: discount rate.²⁸

“We conducted the VOI analysis using a 15-year time horizon and a 3% discount rate.”

VOI indicates value of information.

Item 16

Rationale and description of model: If modeling is used, describe in detail and state why used. *Describe the model structure and justify structural assumptions that have been made.* Report if the model is publicly available and where it can be accessed.

Explanation: EEs often use a range of decision modeling techniques, for example, state-transition models, or decision trees. The choice of technique(s) and model structure will depend on the decision problem and will influence the VOI results.¹⁶ Consequently, it is recommended that the report indicates the modeling technique used, the model structure (eg, health states and allowed transitions), and all structural assumptions. The provided

Table 2. Items included in the CHEERS-VOI checklist.

Topic	Item no.*	Guidance for reporting [†]	Reported in section
Title	1	Identify the study as an economic evaluation and <i>as a VOI analysis</i> , and specify the interventions being compared.	
Abstract	2	Provide a structured summary that highlights context, key methods, results, and alternative analyses.	
Introduction			
Background and objectives	3	Give the context for the study, the study question, and its practical relevance for decision making in policy or practice.	
Methods			
Health economic analysis plan	4	Indicate whether a health economic analysis plan was developed and where it is available.	
Study population	5	Describe characteristics of the study population (such as age range, demographics, socioeconomic, or clinical characteristics). <i>If population-level VOI measures are estimated, describe and justify how the population who benefits from the research was defined, how the incidence and/or prevalence were derived.</i>	
Setting and location	6	Provide relevant contextual information that may influence findings.	
Comparators	7	Describe the interventions or strategies being compared and why they were chosen.	
Perspective	8	State the perspective(s) adopted by the study and why they were chosen.	
Time horizon	9	State the time horizon for the study that is <i>assumed in the economic evaluation</i> and why it is appropriate.	
Discount rate	10	Report the discount rate(s) <i>used in the economic evaluation and for the population-level VOI analysis</i> and the reason why they were chosen.	
Selection of outcomes	11	Describe what outcomes were used as the measure(s) of benefit(s) and harm(s).	
Measurement of outcomes	12	Describe how outcomes used to capture benefit(s) and harm(s) were measured.	
Valuation of outcomes	13	Describe the population and methods used to measure and value outcomes.	
Measurement and valuation of resources and costs	14	Describe how costs were valued.	
Currency, price date, and conversion	15	Report the dates of the estimated resource quantities and unit costs, plus the currency and year of conversion.	
Rationale and description of model	16	If modelling is used, describe in detail and why used. <i>Describe the model structure and justify structural assumptions that have been made.</i> Report if the model is publicly available and where it can be accessed.	
VOI Estimation Methods	S1	(a) Describe the method used to estimate EVPI and any checks used to determine its accuracy. (b) Describe the method used to estimate EVPPI and any checks used to determine its accuracy. (c) Describe the method(s) used to estimate EVSI and any checks used to determine accuracy.	

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Table 2. Continued

Topic	Item no.*	Guidance for reporting [†]	Reported in section
Analytics and assumptions	17	Describe any methods for analyzing or statistically transforming data, any extrapolation methods, and approaches for validating any model used. <i>Make sure to do the following:</i> (a) <i>State and justify the cost-effectiveness threshold(s) chosen.</i> (b) <i>Describe and justify the statistical and methodological choices that were made to estimate parameters (eg, methods of data synthesis and calibration).</i>	
Evidence base	S2	Discuss the potential risk of bias and heterogeneity in the evidence base and describe whether these were adjusted for in the model.	
Characterizing heterogeneity	18	Describe any methods used for estimating how the results of the study vary for subgroups.	
Characterizing distributional effects	19	Describe how impacts are distributed across different individuals or describe adjustments made to reflect priority populations.	
Characterizing uncertainty	20	Describe methods to characterize any sources of uncertainty in the analysis. <i>Discuss the key structural uncertainties and how these have been addressed. For parameters subject to a probabilistic analysis, describe how the distributions reflecting uncertainty were derived, including any dependencies between parameters. For parameters that were not subject to a probabilistic analysis, explain why they are considered fixed and known with certainty.</i>	
Parameters of Interest in VOI analysis	S3	(a) Specify the individual and/or combinations of parameters for which EVPPI has been computed. (b) Specify the individual and/or combinations of parameters for which EVSI has been computed.	
Study Design(s) proposed in VOI analysis	S4	Describe and justify the design of all proposed future research studies and indicate the model parameters that will be updated by these studies.	
Data Generation for EVSI	S5	Describe the assumptions used to generate the study data.	
Costs of Research Studies for ENBS	S6	Provide the costs of research studies proposed in VOI analysis (eg, fixed costs, variable costs, and opportunity costs of allocation), and describe how they were calculated.	
Approach to engagement with patients and others affected by the study	21	Describe any approaches to engage patients or service recipients, the general public, communities, or stakeholders (such as clinicians or payers) in the design of the study.	
Results			
Study parameters	22	Report all analytic inputs (such as values, ranges, and references), including uncertainty or distributional assumptions.	
Summary of main results	23	Report the mean values for the main categories of costs and outcomes of interest, and summarize them in the most appropriate overall measure. <i>Report the values for the VOI measures considered.</i>	
Effect of uncertainty	24	Describe how uncertainty about analytic judgments, inputs, or projections affect findings. Report the effect of choice of discount rate and time horizon, if applicable.	

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Table 2. Continued

Topic	Item no.*	Guidance for reporting [†]	Reported in section
Effect of engagement with patients and others affected by the study	25	Report on any difference patient/service recipient, general public, community, or stakeholder involvement made to the approach or findings of the study.	
Discussion			
Study findings, limitations, generalizability, and current knowledge	26	Report key findings, limitations, and ethical or equity considerations that are not captured and how these could affect patients, policy, or practice.	
Other relevant information			
Source of funding	27	Describe how the study was funded and any role of the funder in the identification, design, conduct, and reporting of the analysis.	
Conflicts of interest	28	Report authors conflicts of interest according to journal or International Committee of Medical Journal Editors requirements.	

CHEERS-VOI indicates Consolidated Health Economic Evaluation Reporting Standards Value of Information; ENBS, expected net benefit of sampling; EVPI, expected value of perfect information; EVPPi, expected value of partial perfect information; EVSI, expected value of sample information; no., number.

*The new items compared with the checklist provided by the CHEERS statement are numbered starting with 1 and proceeded with letter "S."

[†]The modifications of existing items included in the checklist provided by the CHEERS statement are indicated with text in italics.

information is recommended to be sufficiently detailed to ensure transparency and credibility of the model. Guidelines are available to support the evaluation and reporting of structural assumptions for decision-analytic models²⁹ (Box 3²⁸).

BOX 3. Example of item 16: modelling structure.²⁸

"We utilized the Precision Medicine Policy and Treatment (PreEMPT) model, a decision-analytic microsimulation model [citation provided], to provide estimates for our main analysis. Described in detail elsewhere [citation provided], the model simulates the risk of 12 pediatric cancers associated with known autosomal dominant genetic variants. Newborns are assigned a sex and the presence or absence of P/LP variants associated with childhood cancers, and are simulated until death. Each month before age 20 years, individuals are at risk of developing a disease conditional on their variant status. For individuals who develop a disease, the model includes stage at diagnosis, stage-specific survival, and excess treatment-related mortality risks. Further, for all individuals, our model accounts for age-specific other-cause mortality. This study focuses on the pediatric cancer predisposition syndromes associated with TP53 variants and models the benefits of early identification of P/LP TP53 variants. To build this model, we focused on a subset of cancer conditions diagnosed before age of 20 years, including ACC, choroid plexus carcinoma (CPC), OS, and rhabdomyosarcoma (RMS)."

ACC indicates adrenocortical carcinoma; OS, osteosarcoma; P/LP, pathogenic or likely pathogenic.

Item 17

Analytics and assumptions: Describe any methods for analyzing or statistically transforming data, any extrapolation methods, and approaches for validating any model used. *Make sure to:*

- State and justify the cost-effectiveness threshold(s) chosen.
- Describe and justify the statistical and methodological choices that were made to estimate parameters (eg, methods of data synthesis and calibration).

Explanation 17a: VOI analyses require a univariate measure of treatment benefit, whereas most EEs consider 2 outcomes; costs and health.⁷ The cost-effectiveness threshold, which provides a measure of opportunity cost, places these 2 outcomes on the same scale to produce a univariate outcome measure, namely, net monetary or health benefit.^{30,31} Sensitivity analysis to the specific threshold is often used but justification for the specific value or range of values is important because cost-effectiveness thresholds vary widely across jurisdictions and outcomes.³² If the EE is based on a univariate measure of benefit, for example, a cost minimization analysis, then this should be stated (Box 4³³).

BOX 4. Example of item 17a: cost-effectiveness thresholds.³³

"For many countries, a standard willingness-to-pay threshold has been defined or agreed on [citation provided], but not for Germany. We needed to rely on an arbitrary threshold, like a percentage of the gross domestic product (GDP) per capita (46154 Euro in 2016) [citation provided] or similar estimations [citation provided].

The WHO, for example, recommends a 100% GDP threshold to be used. Others have comparatively assessed existing thresholds and have found that for a country like Germany, a realistic threshold would, depending on a number of assumptions and accounting for purchasing-power, range 48-85% of the GDP. To account for this range of possible thresholds, we used (a) a mean of this interval, i.e. 66% of GDP (30461 Euro), in our base case analysis, (b) a minimum of 50% GDP (23077 Euro) and (c) a maximum of 100% GDP (46154 Euro) in two sensitivity analyses [citation provided]."

WHO indicates World Health Organization.

Explanation 17b: Model parameters for the decision-analytic model must be estimated using the information from the evidence base.³⁴ Furthermore, the summary statistics of the parameter (eg, mean and standard error) must appropriately reflect the current state of knowledge. As VOI analysis estimates the value of reducing uncertainty in the model parameters, the parameter estimates and their associated uncertainty influence the VOI results.⁶ There is a range of statistical methodologies that can estimate model parameters, from calculating means and variances to complex methodologies that account for deficiencies in the evidence base such as network meta-analysis,³⁵ digitization of survival curves,³⁶ matched-adjusted indirect comparison,³⁷ and model calibration.^{38–40} It is important to justify why specific estimation methods were used and how they provide estimates for the parameters and their associated uncertainty (Box 5⁴¹).

BOX 5. Example of item 17b: parameter estimation method.⁴¹

"We conducted a network meta-analysis (NMA), multiple-treatment comparison meta-analysis, to simultaneously compare all interventions. Network meta-analysis allows analysts to include not only "direct" comparisons of 2 interventions (e.g., MED v. CABG or CABG v. PCI) but also "indirect" comparisons (e.g., inferring treatment efficacy for MED v. CABG from trials involving MED v. PCI and PCI v. CABG) to obtain combined estimates of the treatment effect [citation provided]. Network meta-analysis provides effectiveness estimates for interventions that have never been compared head-to-head and, by combining direct and indirect effects, increases statistical power to detect differences between treatments [citation provided]. We explored the presence of inconsistency by comparing their direct and indirect effects, qualitatively and quantitatively, with a "split-node" approach [citation provided]."

CABG indicates coronary artery bypass graft surgery; MED, medical therapy; PCI, percutaneous coronary intervention.

Item 20

Characterizing Uncertainty: Describe methods to characterize any sources of uncertainty in the analysis. Discuss key structural uncertainties and how these have been addressed. For parameters subject to a probabilistic analysis, describe how the distributions reflecting uncertainty were derived, including any dependencies between parameters. For parameters that were not subject to a probabilistic analysis, explain why they are considered fixed and known with certainty.

Explanation: Given that the choice of technique(s) and model structure will influence the VOI results,¹⁶ it is recommended to describe all sensitivity analyses performed to evaluate the impact of these structural assumptions. This includes scenario analyses that evaluate the health economic conclusions across a small number of changes to the initial model structure. Further, VOI estimates the value of reducing uncertainty in the model parameters⁶ meaning that VOI is entirely dependent on the statistical uncertainty that is currently assumed for the model parameters. Thus, it is crucial to describe how the input uncertainty is currently characterized.¹⁶ In this item, we distinguish 2 settings that cover all model parameters. First, reports are recommended to discuss the model parameters that are subject to statistical uncertainty and how their uncertainty distributions are derived. This will link to the reporting of item 17, particularly when analyses are undertaken in a Bayesian

framework.⁴² This item builds on item 17 to clarify the distributional assumptions for the model parameters and the reason these distributions were chosen. It is crucial that all input uncertainty is correctly parameterized by assigning appropriate probability distribution, including any dependency between parameters by assigning a joint, correlated probability distribution. Second, for model parameters that are not subject to uncertainty, the fixed value chosen for these parameters should be clarified. It is also essential that the report provides a discussion on why these parameters are assumed to be known with certainty because any parameters that are assumed fixed will have no associated VOI (Box 6^{43–45}). In some settings, these fixed parameters could be explored in a sensitivity analysis, which would also be specified as a structural assumption in item 16.

BOX 6. Example of item 20: characterizing uncertainty.^{43–45}

STRUCTURAL UNCERTAINTY: "Two alternative model specifications were analyzed to explore key structural uncertainties:

1. *Sensitivity analysis on the chemotherapy effect:* The assumption of a predictive treatment effect was challenged in a sensitivity analysis that assumed constant relative benefit from chemotherapy across all risk groups on the basis of the Oxford Overview meta-analysis applied over a 5-year period [citation provided].
2. *Sensitivity analysis on survival after recurrence:* In the base-case analysis, the survival after distant recurrence was assumed to be constant across groups. In a sensitivity analysis, survival after recurrence was assumed to depend on whether patients had previously received adjuvant chemotherapy."

PARAMETER UNCERTAINTY: "(a) Probabilistic input parameters of the model included transition probabilities between disease states, utilities associated with the disease states, number of hospitalizations (with and without ER visits), costs associated with the disease states (resource utilization including specialist/ general practitioner/ nurse contacts), and initial distribution of patients per NYHA group. Uncertainty around these parameters was considered simultaneously and each parameter was sampled from the corresponding distribution, independently. This was an important assumption in order to count the total costs and QALYs generated within all three intervention arms. We sampled the transition probabilities between disease states from Dirichlet distribution [citation provided], and used beta distribution for sampling utilities, and uniform distribution for sampling the installment/ equipment and service fee for HTM. Second-order Monte Carlo simulations (i.e., considering parameter uncertainty) were undertaken in which values were randomly drawn from these distributions.

(b) Daily costs of all medications, including LCZ696, were based on the Z-index [citation provided] and further calculated applying the following formula from the 2010 Dutch costing manual [citation provided] that was valid at the time the analyses were done: *Medication costs = pharmacists purchase price (Z-index) - clawback (8.3%) + VAT (6%) + pharmacy dispensing fee*. The daily cost of LCZ696 (based on the recommended dose) was €4.95 for 200 mg twice a day (which corresponds with sacubitril/valsartan 97 mg/103 mg twice a day). Applying the formula above, the daily cost of LCZ696 used in the model was €4.83. The daily cost of enalapril (based on the target dose as defined in the PARADIGM-HF trial protocol) was €0.14 for 10 mg twice a day."

ER indicates emergency room; HTM, Home Telemonitoring; LCZ696, angiotensin receptor neprilysin inhibitor LCZ696; NYHA, New York Heart Association; QALY, quality-adjusted life year.

New Items

The items elaborated on below represent items that were not part of the core CHEERS statement but are recommended to be included when VOI is performed alongside EE.

Item S1

VOI Estimation Methods:

- Describe the method used to estimate EVPI and any checks used to determine its accuracy.
- Describe the method used to estimate EVPPI and any checks used to determine its accuracy.
- Describe the method(s) used to estimate EVSI and any checks used to determine accuracy.

Item S1a. EVPI Estimation Method: Describe the method used to estimate EVPI and any checks used to determine its accuracy.

Explanation: EVPI can rarely be calculated analytically⁴⁶ and must usually be estimated through simulation.⁹ It is important to clarify how these simulations were undertaken, including the number of simulations used in EVPI estimation, the accuracy of the simulation-based estimate, for example, the standard error, and any validation checks that were used (Box 7⁴⁷⁻⁵⁰).

BOX 7. Example of item S1a: EVPI estimation method.⁴⁷⁻⁵⁰

METHOD: "We assessed the expected value of perfect information (EVPI) by averaging the maximum NMB over the joint distribution of all the parameters in the model."
 METHOD: "The number of iterations for the EVPI analysis and the one-level EVPPI analysis was set to 10,000."
 METHOD: "We used a no-modeling approach to VOI. First, we used published data on time to symptom resolution presented in the form of survival curves to recover the individual level data for time to symptom resolution under the two treatment arms. [...] We estimated the individual level value of perfect information by bootstrap sampling out of the distribution of average benefits and costs and averaging the net benefit of azithromycin over amoxicillin/clavulanate whenever azithromycin had a positive net benefit."
 CHECKS: "We additionally used Excel to validate our calculations with respect to Net Benefit and EVPI."

EVPI indicates expected value of perfect information; NMB, net monetary benefit; VOI, value of information.

Item S1b. EVPPI Estimation Method: Describe the method used to estimate EVPPI and any checks used to determine its accuracy.

Explanation: There are several methods available to estimate EVPPI,⁵¹ including nested Monte Carlo methods,⁴⁶ nonparametric regression methods,⁵² and multilevel Monte Carlo methods.⁵³ The method(s) used to estimate EVPPI are recommended to be presented alongside information such as simulation size or regression method chosen. Furthermore, the use of accuracy checks, such as calculating the SD,⁵² bias estimation,⁵⁴ or residual checking,⁵⁵ are recommended to be clarified (Box 8^{50,56,57}).

BOX 8. Example of item S1b: EVPPI estimation method.^{50,56,57}

METHOD: "We used multilevel Monte Carlo to estimate per-person expected value of partial perfect information (EVPPI)."
 CHECKS: "Standard errors of the conventional EVPPI estimates were calculated by taking the standard deviation of 10,000 bootstrapped EVPPIs."
 CHECKS: "The number of iterations for the EVPI analysis and the one-level EVPPI analysis was set to 10,000. For the two-level algorithms, the number of inner loops was set to 1000, whereas the number of outer loops was set to 500. The number of outer loops was smaller than the number of inner loops because, in this specific analysis, there is less variation in the outer loop than in the inner loop."

EVPI indicates expected value of perfect information.

Item S1c. EVSI Estimation Method: Describe the method(s) used to estimate EVSI and any checks used to determine accuracy.

Explanation: There is a range of methods available to estimate EVSI,⁵⁸ which require different inputs.⁵⁹ Thus, it is important to describe which method has been used and report any method specific inputs required to reproduce the EVSI estimate. Accuracy checks will also vary depending on the model, for example, nonparametric regression-based methods require residual plotting,⁶⁰ whereas some methods provide an estimate of the standard error of the EVSI estimate^{60,61} (Box 9^{62,63}).

BOX 9. Example of item S1c: EVSI estimation method.^{62,63}

METHOD: "We use the non-parametric regression approach [to calculate EVSI], as this does not require the existence of conjugate distributions or parametric assumptions."
 CHECKS: "We computed 95% intervals for the GAM estimator by sampling 2,000 values from a multivariate Normal distribution of the GAM coefficients, as described in an appendix of the article by Strong et al. [citation provided]"

GAM indicates generalized additive model.

Item S2

Evidence Base: Discuss the potential risk of bias and heterogeneity in the evidence base and describe whether these were adjusted for in the model.

Explanation: VOI analysis estimates the value of gaining additional information to support medical decision making.⁶ Implicitly, VOI estimations assume that this information is added to the current evidence base, meaning that the current evidence directly affects the VOI.¹⁶ The current evidence may be extracted from (1) previous studies, (2) individual patient data, or (3) experts through structured expert elicitation procedures.⁶⁴ It is important to clarify (1) which studies are included in the evidence base and the limitations of these studies and/or (2) how expert opinion was formally extracted.⁶⁵ Heterogeneity and bias in the previous studies or elicitation will affect the VOI and are recommended to

be described in line with the recommendations for the assessment of heterogeneity and risk of bias in decision-analytic modeling^{34,65} (Box 10⁴⁷).

BOX 10. Example of item S2: evidence base.⁴⁷

"[...] comparative evidence from the Silkie before and after study was used. The need for re-grafting on low-friction bedding was a prospectively collected study outcome and the relative risk of low-friction versus standard was informed by the before and after comparison. Because the lack of concealed randomization can lead to problems of comparability because of selection biases and potential confounders [citation provided] and because before/after designs are by definition prone to history bias [citation provided]. [...] ideally, data on utilities would be routinely collected in the population with burn injury to allow for service evaluation and monitoring. A population-level sample would greatly increase the precision around the estimate of expected QALYs. However, measurement of health-related QoL is not routine in the UK burn populations, so the utility estimates for patients who did and did not undergo re-graft surgeries were identified from the perspective (intervention arm) of Silkie study."

QALY indicates quality-adjusted life-year; UK, United Kingdom.

Item S3

Parameter of Interest in VOI analysis:

- Specify the individual and/or combinations of parameters for which EVPPI has been computed.
- Specify the individual and/or combinations of parameters for which EVSI has been computed.

Item S3a. Parameters of Interest in EVPPI: Specify the individual and/or combinations of parameters for which EVPPI has been computed.

Explanation: It is crucial to specify which subsets have been considered and to clearly identify which EVPPI values are associated with which subsets. Justification for why certain parameter subsets have been considered may also be provided (Box 11⁶⁶).

BOX 11. Example of item S3a: parameters of interest in EVPPI.⁶⁶

"Parameter subsets considered were all costs, only event costs, only state costs, all utilities, only event utilities, only state utilities, treatment switching probabilities, impact of past events on future risks, baseline (warfarin) log hazards, and DOAC treatment effects relative to warfarin. In addition, we calculated EVPPI for the relative effects parameters that would be informed by a two-arm RCT comparing the top two cost-effective treatments, a three-arm RCT comparing the top three cost-effective treatments, and so on for four-arm and five-arm RCTs."

DOAC indicates directly acting (or non-vitamin K antagonist) oral anticoagulants; EVPPI, expected value of partial perfect information; RCT, randomized clinical trial.

Item S3b. Parameters of Interest in EVSI: Specify the individual and/or combinations of parameters for which EVSI has been computed.

Explanation: EVSI estimates the value of reducing decision uncertainty by gathering additional evidence for specific parameters or subsets of parameters used to inform the decision-analytic model.⁹ It is crucial to specify which parameters or subsets of parameters have been considered and to clearly identify which EVSI values are associated with which subsets. Justification for why certain parameter subsets have been considered may also be provided (Box 12²⁸).

Item S4

Study Design(s) proposed in VOI analysis: Describe and justify the design of all proposed future research studies and indicate the model parameters that will be updated by these studies.

Explanation: EVSI estimates the value of undertaking research studies with specified designs.¹¹ It is crucial, therefore, to clearly articulate the design of the studies that are considered. It is recommended that this includes the research question (eg, population, interventions, control, and outcome) and the links to model parameter(s) will be updated by the study, reported in item S3b. The study type(s), for example, randomized controlled trial, observational study, administrative data study, and questionnaire, is also recommended to be clarified. Finally, other key design considerations, for example, sample size, length of follow-up, and treatment switching, should be discussed (Box 13⁶⁷).

Item S5

Data Generation: Describe the assumptions used to generate the study data.

Explanation: EVSI estimations typically require the simulation of future data from the studies of interest.⁶⁸ These simulated data should represent individual-level heterogeneity in the outcomes, conditional on the model parameters of interest that were identified in item 18. It is crucial to specify any distributional assumptions that were made to simulate the data. Information on how individual-level variability is defined and the values of inputs that parameterize this variability are recommended to be provided. This applies also to decision-analytic models that do not consist of individual-level variability parameters, such as a cohort level model⁶⁹ (Box 14⁵⁸).

Item S6

Costs of Research Studies for ENBS: Provide the costs of research studies proposed in VOI analysis (eg, fixed costs, variable costs, and opportunity costs of allocation), and describe how they were calculated.

Explanation: ENBS is the difference between the population-level EVSI and the cost of the proposed study/ies.¹² Thus, it is critical to specify the assumed cost of research used to compute ENBS. It is recommended to identify which research costs have been considered and how their values have been estimated. Research costs may include fixed costs and per-patient costs, which will vary depending on the study design and are recommended to be justified.¹² In small patient populations, opportunity costs should also be considered because individuals who receive their treatment before the information is available will not benefit from the evidence.¹² Finally, in interventional studies, costs can include the opportunity cost of individuals receiving the nonoptimal treatment as part of the research strategy¹² (Box 15⁷⁰).

BOX 12. Example of item S3b: parameters of interest in EVSI.²⁸

"Table XX. Model parameters and assumptions included in each of the proposed research studies

Proposed research study	Research study population	Research study outcomes	Research study type	Research study costs
Research Study 1	Individuals with ACC	1. <i>TP53</i> P/LP prevalence among individuals with ACC 2. Proportion diagnosed with localized ACC ^a 3. 5-year survival rate for localized ACC ^b 4. 5-year survival rate for advanced ACC ^c 5. Cumulative ACC risk by age 20	Retrospective observational registry study	Fixed cost: \$90,000 Cost per patient: \$60
[...]				
Research Study 5	Individuals without any of the cancers associated with <i>TP53</i> variants (general population)	1. <i>TP53</i> P/LP prevalence in general population	Retrospective observational registry study	Fixed cost: \$90,000 Cost per patient: \$60

^aWe defined advanced disease as regional, distant and unstaged tumors. ACC, adrenocortical carcinoma; P, pathogenic variant; pop, population; LP, likely pathogenic variant.

^bP/LP variants associated with cancer cases diagnosed before age 5 only.

^cBased on data from the SEER Program.

^dBased on age-specific SEER incidence rates between ages 0-19 (<1, 1-4, 5-9, 10-14, 15-19 years old).

SEER indicates Surveillance, Epidemiology, and End Results Program.

Discussion and Concluding Remarks

Standardized reporting of VOI analyses is needed to improve the interpretation and comparison of studies and ensure the analyses provide accurate research recommendations. Given the increasing use of VOI, we developed the CHEERS-VOI checklist, which adapted the CHEERS statement¹³ to studies that perform a VOI analysis alongside EE. We used similar methodology to the development of the CHEERS statement to incorporate the views of international and multidisciplinary experts in VOI analysis and EE.

The length and comprehensiveness of the CHEERS-VOI is similar to the CHEERS statement. Thus, when EE includes a VOI analysis, the CHEERS-VOI checklist should be used for reporting purposes instead of the CHEERS statement.

The CHEERS-VOI checklist contains all 28 items included in the CHEERS statement, 7 of which required modification, and 6 new reporting items. All these 33 items included in the CHEERS-VOI checklist represent the minimum, essential reporting criteria for EEs that include a VOI analysis. The steering committee considered inclusion additional items (eg, imperfect implementation in VOI estimation and software used) but only the reporting items included in the final CHEERS-VOI checklist were considered as

BOX 13. Example of item S4: study design.⁶⁷

"We studied a hypothetical randomized-controlled trial (RCT) that aimed to investigate the effect size of an alcohol intervention for HIV infected patients in East Africa with following assumptions: (1) Participants will be equally randomized to an intervention arm and a control arm; (2) Participant eligibility criteria are hazardous or binge drinkers (score ≥ 3 on the Alcohol Use Disorders Identification Test (AUDIT-C), a brief screening for heavy drinking and/or active alcohol abuse or dependence [citation provided]) and being ART-eligible or ART-initiated in the past 12 months; (3) The trial could be funded for 6–36 months; [...]

Note that the primary interest of this work was to investigate evidence necessary to reduce uncertainty regarding the effect size of an alcohol intervention. The effect size was measured as the relative risk reduction of unhealthy alcohol consumption."

ART indicates antiretroviral treatment.

BOX 14. Example of item S5: data generation.⁵⁸

"EVSI is computed for a study that investigates the QoL weights for patients who remain on treatment without any adverse events and for patients who withdraw from the first-line treatment due to lack of efficacy. The individual level variability in these two QoL weights is modelled, for simplicity, as independent beta sampling distributions although the assumption of independence may be invalid [citation provided]. The population level mean QoL weight, i.e., the mean of the sampling distribution for the QoL weights, is defined as the value of those two health states in the Markov model. The standard deviations of the individual level sampling distributions are then set equal to 0.3, for patients who remain on treatment, and 0.31, for patients who withdraw due to lack of efficacy [citation provided]."

QoL indicates quality of life.

BOX 15. Example of item S6: study costs for ENBS.⁷⁰

"The total cost of a future trial should include fixed costs (e.g. salaries), variable costs (i.e. per patient recruited), and opportunity costs (i.e. benefits forgone) incurred by patients who receive the inferior intervention while the trial is performed. [citation provided] [...] The total cost for a future study was estimated to be AUD120,000 of fixed costs and AUD150/patient in variable cost."

AUD indicates Australian Dollar; ENBS, expected net benefit of sampling.

minimal, essential information to be reported. Importantly, the CHEERS-VOI checklist can be used to assess the quality of reporting of a study but not the quality of its conduct. Further, the CHEERS-VOI checklist is not intended to provide an explanation of VOI or guidance on how to conduct and interpret VOI. Previous work on the role of VOI and VOI methods, including the 2 foundational reports published by the ISPOR VOI Task Force,^{17,18} as well as other previous work,^{7,15,51,58,59} may be used alongside the CHEERS-VOI checklist.

Limitations in our approach include using a quorum of only 5 steering committee members during our consensus meetings, which ensured feasibility but may have meant limited input was available at the consensus stage. Further, we acknowledge that our study lacked patient engagement. Finally, although we aimed to secure diversity in the experience represented in our study and involved a large number of researchers and other stakeholders with expertise in VOI and EE, we acknowledge that the current work reflects the experience and perspective of the involved study participants. Future work should focus on assessing the impact of CHEERS-VOI on the reporting of VOI analyses and potentially revisiting and extending the checklist as VOI analysis methods continue to evolve.

The present explanation and elaboration report aims to aid users to appropriately apply the CHEERS-VOI checklist when their analysis includes a VOI analysis alongside EE. The CHEERS-VOI checklist is primarily intended for researchers, analysts, and other stakeholders who either report or assess an EE with a VOI analysis, which is aimed to be published in peer-reviewed journals or in form of other reports or submitted for health technology assessment.

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Accepted for Publication: June 20, 2023

Published Online: August 14, 2023

doi: <https://doi.org/10.1016/j.jval.2023.06.014>

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Critical revision of the article for important intellectual content: Kunst, Siu, Drummond, Grimm, Grutters, Husereau, Koffijberg, Rothery, Wilson, Heath

Obtaining funding: Heath

Administrative, technical, or logistic support: Kunst, Siu, Heath

Supervision: Kunst, Heath

Conflict of Interest Disclosures: The authors reported no conflicts of interest. Drs Drummond, Grimm, and Rothery are editors for *Value in Health* and had no role in the peer-review process of this article.

Funding/Support: Dr Heath is supported by a Canada Research Chair in Statistical Trial Design and funded by the Discovery Grant Program of the Natural Sciences and Engineering Research Council of Canada (RGPIN-2021-03366). Ms Siu was supported by the Government of Canada's New Frontiers in Research Fund (NFRF), (NFRFE-2020-00749).

Role of the Funder/Sponsor: The funder had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the article; or decision to submit the article for publication.

Acknowledgment: The CHEERS-VOI steering committee thank Jingxian Lan who reviewed the titles and abstracts for our literature review and all ConVOI members for their feedback on the initial list of candidate items. The CHEERS-VOI steering committee also thank all the participants in the Delphi study. Their expertise was vital to the development of this checklist. The names of those who allowed acknowledgment are listed below in alphabetical order:

Eline Aas, Lusine Abraham, Anirban Basu, Doug Coyle, Elisabeth Fenwick, Sean Gavan, Salah Ghabri, David Glynn, Jason Guertin, Christopher Jackson, Jonathan Karnon, Saskia Knies, Ka Keat Lim, Robert Brett McQueen, Mike Paulden, Bram Ramaekers, Moshen Sadatsafavi, Paul Scuffham, Lotte Steuten, Kednapa Thavorn, Haitham Tuffaha, Mathyn Vervaaert, Rick Vreman, Nicky Welton

Ethics Approval and Consent to Participate: The Research Ethics Board at the Hospital for Sick Children deemed that this study did not require ethical approval. All participants were consented to participate in the registration survey.

Availability of Data and Material: The complete Delphi summary document is available from the authors upon reasonable request.

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